

Linear Programming Study Set

1/21

Question 1. *Imagine you are the operator of a clothing factory which only produces two products: sweater and vest. Each sweater can be sold at \$75 and takes 1.5 hr and 2m^2 of cloth to be made. Each vest can be sold at \$60 and takes 1 hr and 1m^2 of cloth to be made. You have 600 labor hours and 900m^2 of cloth in total. How do you maximize your profit given the resources? It costs \$20 to produce both sweaters and vests. Please formulate the problem into an LP and solve it. For this question, we can assume having non-integer numbers of vests or sweaters are fine.*

Correct Answer 1.

Decision variables:

x_1 : number of vests to produce

x_2 : number of sweaters to produce

Objective Function

The objective is to maximize the profit, Z :

$$Z = (60 - 20)x_1 + (75 - 20)x_2$$

Constraints

1. Time Constraint

$$x_1 + 1.5x_2 \leq 600$$

2. Resource Constraint

$$x_1 + 2x_2 \leq 900$$

3. Non-negativity Constraints

$$x_1, x_2 \geq 0$$

Linear Programming Study Set

2/21

Question 2. (Modified from Bertsimas's Exercise 1.9 ^a)

The Challenge Facing UrbanEd Consulting and the City of Harmonyville

UrbanEd Consulting, a firm specialized in optimizing educational resources, has been hired by the local government of Harmonyville to tackle a pressing logistical issue that's been the talk of every PTA meeting in recent months. Harmonyville is a city comprising five diverse neighborhoods: Parkside, Lakeview, Forest Glen, Sunnydale, and Riverbank. The city is home to three public schools that serve grades K-5: Harmony Elementary, New Horizon School, and Pioneer Academy.

The neighborhoods vary in their student populations, and each school has its own capacity limitations for each grade level. Meanwhile, parents are concerned about the distance their children have to travel to school, especially with rising fuel costs and concerns about environmental sustainability.

Local agencies have provided UrbanEd with the following essential information:

1. **Student Population:** *Each neighborhood has provided the number of students in each grade level (K-5), obtained from the most recent census.*
2. **School Capacity:** *Each school has listed its maximum capacity for each grade level.*
3. **Distance Metrics:** *The Department of Transportation has calculated the distance from each neighborhood to each of the three schools.*

UrbanEd's task is to develop a school assignment plan that places all students in a school while minimizing the total distance traveled by all students. This will not only reduce costs for families but also lower the community's carbon footprint.

—
P.S. You may ignore the fact that numbers of students must be integer.

^aBertsimas, D., Tsitsiklis, J. (1998). Introduction to Linear Optimization. In Athena Scientific.

Linear Programming Study Set

3/21

Correct Answer 2. To address this challenge, UrbanEd decides to formulate it as a linear programming problem.

Parameters:

- N : Set of neighborhoods, $N = \{\text{Parkside, Lakeview, Forest Glen, Sunnydale, Riverbank}\}$.
- S : Set of schools, $S = \{\text{Harmony Elementary, New Horizon School, Pioneer Academy}\}$.
- G : Set of grade levels, $G = \{K, 1, 2, 3, 4, 5\}$.
- $d_{n,s}$: Distance from neighborhood $n \in N$ to school $s \in S$.
- $p_{n,g}$: Population (number of students) in neighborhood $n \in N$ for grade $g \in G$.
- $c_{s,g}$: Maximum capacity for grade $g \in G$ in school $s \in S$.

Decision Variables:

Let $x_{n,s,g}$ be the number of students from neighborhood n attending school s in grade g .

Objective Function:

Minimize the total distance traveled by all students:

$$\text{Minimize } Z = \sum_{n \in N} \sum_{s \in S} \sum_{g \in G} d_{n,s} \times x_{n,s,g}$$

Constraints:

1. Student Assignment Constraints: All students must be assigned to some school.

$$\sum_{s \in S} x_{n,s,g} = p_{n,g} \quad \forall n \in N, g \in G$$

2. School Capacity Constraints: The number of students assigned to each grade in each school cannot exceed the school's capacity for that grade.

$$\sum_{n \in N} x_{n,s,g} \leq c_{s,g} \quad \forall s \in S, g \in G$$

3. Non-negativity Constraints: The number of students assigned must be non-negative. (Also, we can consider integrality here.)

$$x_{n,s,g} \geq 0 \quad \forall n \in N, s \in S, g \in G$$

UrbanEd will then use linear programming tools to solve this problem, providing the city of Harmonyville with the most efficient school assignment plan.

Linear Programming Study Set

4/21

```
from gurobipy import Model, GRB

# Initialize model
m = Model("school_assignment")

# Decision Variables
x = {}
for n in neighborhoods:
    for s in schools:
        for g in grades:
            x[n, s, g] = m.addVar(vtype=GRB.INTEGER, name=f"x_{n}_{s}_{g}")

# Objective Function
m.setObjective(sum(distances[n, s] * x[n, s, g] for n in neighborhoods for
    s in schools for g in grades), GRB.MINIMIZE)

# Constraints
for s in schools:
    for g in grades:
        m.addConstr(sum(x[n, s, g] for n in neighborhoods) <= capacities[s,
            g], f"Capacity_{s}_{g}")

for n in neighborhoods:
    for g in grades:
        m.addConstr(sum(x[n, s, g] for s in schools) == students[n, g], f"
            Demand_{n}_{g}")

# Solve Model
m.optimize()

# Display Results
if m.status == GRB.OPTIMAL:
    for n in neighborhoods:
        for s in schools:
            for g in grades:
                if x[n, s, g].x > 0:
                    print(f"Number_of_students_from_{n}_assigned_to_{s}_in_
                        grade_{g}:_{x[n,s,g].x}")
    print(f"Total_distance_traveled_by_all_students:_{m.objVal}")
```

Linear Programming Study Set

5/21

Question 3. (Taha's Exercise 2.5 ^a) Determine the feasible region for each of the following independent constraints, given that $x_1, x_2 \geq 0$.

1. $-3x_1 + x_2 \leq 6$

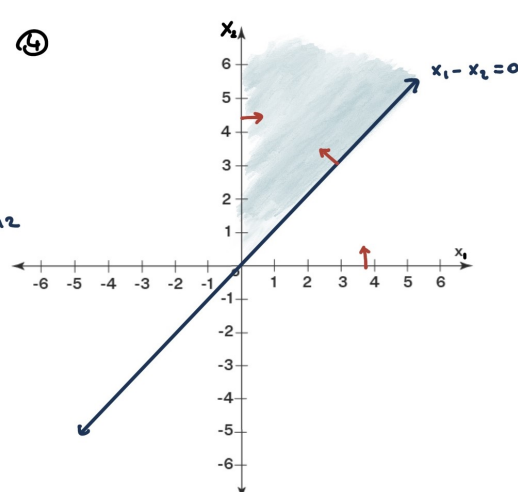
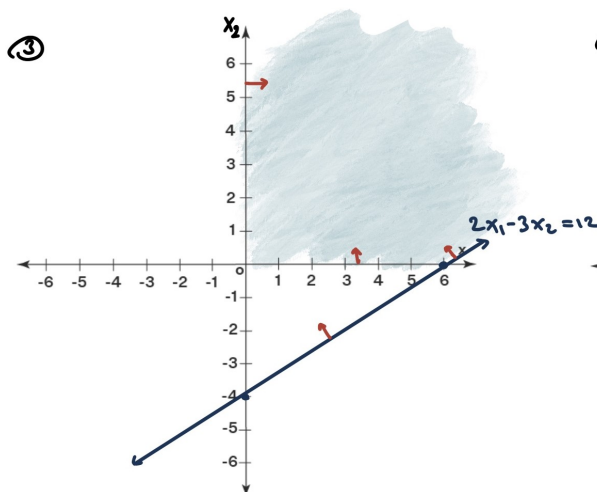
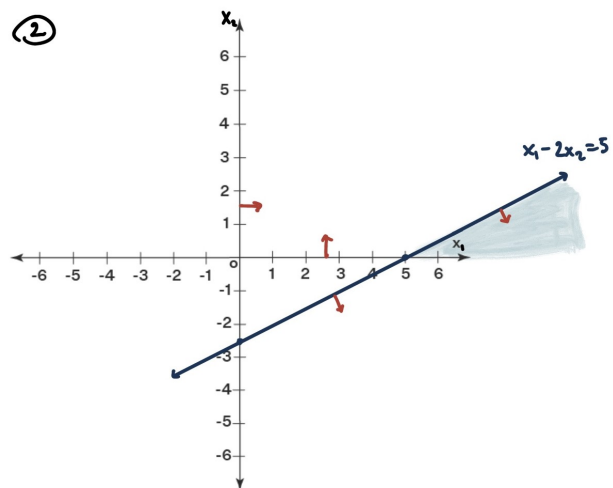
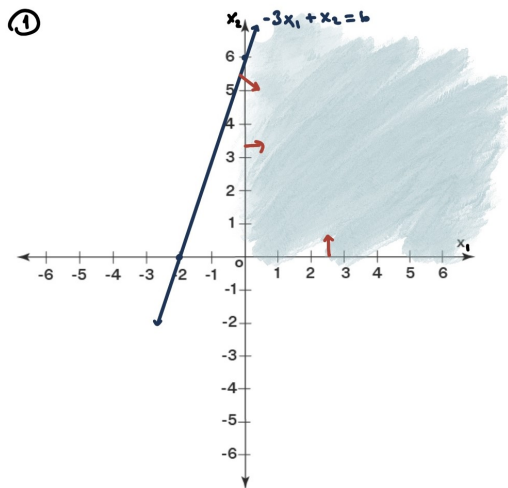
2. $x_1 - 2x_2 \geq 5$

3. $2x_1 - 3x_2 \leq 12$

4. $x_1 - x_2 \leq 0$

^aTaha, H. A. (2006). Operations Research: An Introduction (8th Edition). Prentice-Hall, Inc.

Correct Answer 3.



Linear Programming Study Set

6/21

Question 4. (Taha's Exercise 2.43 ^a)

In anticipation of the immense college expenses, Arjun and Wang started an annual investment program on their child's eighth birthday that will last until the eighteenth birthday.

To avoid unpleasant surprises, they want to invest the money safely in the following options: insured savings with 7.5% annual yield, 6-year government bonds that yield 7.9% and have a current market price equal to 98% of face value, and 9-year municipal bonds yielding 8.5% and having a current market price of 1.02 of face value. Their investment plan is given in Table 1. Can you help Arjun and Wang find the optimal investment plan?

^aTaha, H. A. (2006). Operations Research: An Introduction (8th Edition). Prentice-Hall, Inc.

Table 1: Arjun and Wang's investment plan

Year	1	2	3	4	5	6	7	8	9	10
Amount (\$)	2000	2000	2500	2500	3000	3500	3500	4000	4000	5000

Linear Programming Study Set

7/21

Correct Answer 4.

In this context, we can define our objective function as the sum of the future values of all investments, and we want to maximize this objective.

Decision Variables:

Let x_i be the amount invested in the insured savings in the i^{th} year.

Let y_i be the amount invested in the 6-year government bonds in the i^{th} year.

Let z_i be the amount invested in the 9-year municipal bonds in the i^{th} year.

Objective Function

Our objective is to maximize the sum of the future values of all the investments, which is the sum of the returns from each investment option:

$$\text{Maximize } Z = \sum_{i=1}^{10} x_i(1 + 0.075)^{(11-i)} + \sum_{i=1}^5 y_i(1 + 0.079)^{(11-i)} + \sum_{i=1}^2 z_i(1 + 0.085)^{(11-i)}$$

Constraints

1. For each year, the sum of investments in all options should be equal to the amount available for that year, considering the face values.

$$\begin{aligned}x_1 + 0.980 \times y_1 + 1.02 \times z_1 &= 2000 \\x_2 + 0.980 \times y_2 + 1.02 \times z_2 &= 2000 \\x_3 + 0.980 \times y_3 + 1.02 \times z_3 &= 2500 \\x_4 + 0.980 \times y_4 + 1.02 \times z_4 &= 2500 \\x_5 + 0.980 \times y_5 + 1.02 \times z_5 &= 3000 \\x_6 + 0.980 \times y_6 + 1.02 \times z_6 &= 3500 \\x_7 + 0.980 \times y_7 + 1.02 \times z_7 &= 3500 \\x_8 + 0.980 \times y_8 + 1.02 \times z_8 &= 4000 \\x_9 + 0.980 \times y_9 + 1.02 \times z_9 &= 4000 \\x_{10} + 0.980 \times y_{10} + 1.02 \times z_{10} &= 5000\end{aligned}$$

2. Time constraints: You can only invest in 6-year government bonds up to Year 5, and in 9-year municipal bonds up to Year 2.

$$\begin{aligned}y_6 = y_7 = y_8 = y_9 = y_{10} &= 0 \\z_3 = z_4 = z_5 = z_6 = z_7 = z_8 = z_9 = z_{10} &= 0\end{aligned}$$

3. Non-negativity constraints: Investments can't be negative.

$$x_i, y_i, z_i \geq 0 \text{ for all } i$$

Linear Programming Study Set

8/21

Question 5. *You're working as a production engineer at a company that produces different grades of jet fuels and one day you receive this e-mail from your manager:*

"Hey,

We've got a bit of a situation on our hands, and we need your expertise to optimize our fuel production. As you know, we're producing two grades of jet fuel: F1 and F2. These fuels are blended from four types of raw gasoline: A, B, C, and D.

For F1, the blend is straightforward: a 1:1:2:4 ratio of A to B to C to D. For F2, we're looking at a 2:2:1:3 ratio of A, B, C, and D. I trust you've got all the technical specs on these.

Now, here's the catch. Our daily supply limits are tightening up. We've got only 1000 barrels of A, 1200 of B, 900 of C, and 1500 of D available per day. The raw material costs are also a concern: A costs us \$120 a barrel, B is at \$90, C at \$100, and D is the most expensive at \$150 a barrel.

On the sales side, we're doing \$200 a barrel for F1 and \$250 a barrel for F2. But the market won't take anything less than 200 barrels of F1 and 400 barrels of F2 a day.

We need to meet market demands while optimizing for costs and supply limitations. Time is of the essence; let's maximize our profits. Can you figure out the best daily production mix for these two fuels? "

How would you solve this problem?

Linear Programming Study Set

9/21

Correct Answer 5. Let's define x_1 and x_2 as the barrels of F1 and F2 produced per day, respectively. The decision variables a, b, c , and d represent the barrels of A, B, C, and D used per day, respectively.

Blending Ratios for F1 and F2 are given below:

1. For F1: $a = \frac{x_1}{8}, b = \frac{x_1}{8}, c = \frac{2x_1}{8}, d = \frac{4x_1}{8}$

2. For F2: $a = \frac{2x_2}{8}, b = \frac{2x_2}{8}, c = \frac{x_2}{8}, d = \frac{3x_2}{8}$

These can be translated to:

$$a = \frac{x_1}{8} + \frac{2x_2}{8}, \quad b = \frac{x_1}{8} + \frac{2x_2}{8}, \quad c = \frac{2x_1}{8} + \frac{x_2}{8}, \quad d = \frac{4x_1}{8} + \frac{3x_2}{8}.$$

Objective Function

The objective is to maximize the profit, Z :

$$Z = 200x_1 + 250x_2 - (120a + 90b + 100c + 150d)$$

Constraints

1. Supply Constraints

$$0 \leq a \leq 1000$$

$$0 \leq b \leq 1200$$

$$0 \leq c \leq 900$$

$$0 \leq d \leq 1500$$

2. Demand Constraints

$$x_1 \geq 200$$

$$x_2 \geq 400$$

3. Non-negativity Constraints

$$a, b, c, d, x_1, x_2 \geq 0$$

To remove a, b, c , and d from the LP model, we can substitute their expressions directly into the objective function and constraints. This will give us an LP model purely in terms of x_1 and x_2 .

Linear Programming Study Set

10/21

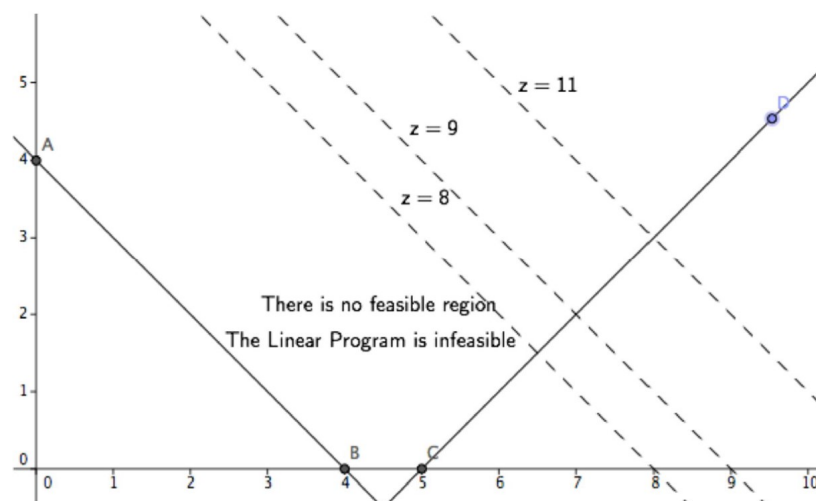
Question 6. Use graphical method to solve the following problem:

$$\begin{aligned} \max \quad & Z = x_1 + x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 4 \\ & x_1 - x_2 \geq 5 \\ & x_1, x_2 \geq 0 \end{aligned}$$

CorrectAnswer 6.

Equation $x_1 + x_2 = 4$ is represented by straight line AB.

CD represents $x_1 - x_2 = 5$.



Linear Programming Study Set

11/21

Question 7. Lee Chemical manufactures three chemicals: Sodium hydroxide, Nitrogen, and Propylene. These chemicals are produced via two production processes: 1 and 2. Running process 1 for an hour costs \$4 and yields 3 liters of Sodium hydroxide, 1 Nitrogen, and 1 of Propylene. Running process 2 for an hour costs \$1 and produces 1 liters of Sodium hydroxide, and 1 of Nitrogen. To meet customer demands, at least 10 liters of Sodium hydroxide, 5 of Nitrogen, and 3 of Propylene must be produced daily. Graphically determine a daily production plan that minimizes the cost of meeting Lee chemical's daily demands.

Correct Answer 7.

Let, x_1 be the number of hours of Process 1 and x_2 be the number of hours of Process 2.

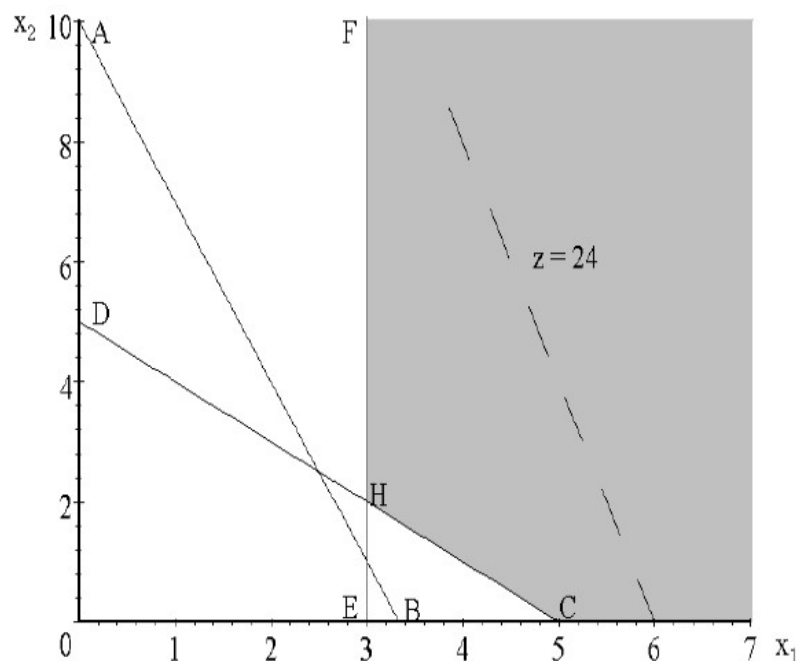
Then the appropriate LP is:

$$\begin{aligned} \min \quad & Z = 4x_1 + x_2 \\ \text{s.t.} \quad & 3x_1 + x_2 \geq 10 \quad (\text{Constraint - Sodium hydroxide}) \\ & x_1 + x_2 \geq 5 \quad (\text{Constraint - Nitrogen}) \\ & x_1 \geq 3 \quad (\text{Constraint - Propylene}) \\ & x_1, x_2 \geq 0 \end{aligned}$$

In the Figure below AB is $3x_1 + x_2 = 10$. CD is $x_1 + x_2 = 5$. EF is $x_1 = 3$. The feasible region is shaded.

Dotted line is isocost line $4x_1 + x_2 = 24$.

Moving isocost line down to left we see that H (where B and C constraints intersect) is optimal. Thus, the optimal solution to this LP is $x_1 = 3, x_2 = 2, z = 4(3) + 2 = \14 .



Linear Programming Study Set

12/21

Question 8. (Modified from Troutt, M. D. (1983) ^a)

A small university offers two majors: Engineering and Arts. The university has both in-state and out-of-state students. The objective is to determine the optimal fixed and variable tuition fees for each category of students to maximize the total income, considering various constraints including lower and upper bounds on fees and maintaining average tuition below a certain limit. Below are the estimated demand, and certain limitations that are given by the school.

Parameters:

- $E_i = 100$: Expected number of in-state Engineering students.
- $A_i = 150$: Expected number of in-state Arts students.
- $E_o = 150$: Expected number of out-of-state Engineering students.
- $A_o = 100$: Expected number of out-of-state Arts students.
- $C = 15$: Average number of credits taken by a student.
- Capacity: 500 students.
- Minimum Income Requirement for the University: \$10,000,000.
- Upper bound on average tuition for all students: \$25,000.
- Upper bound of fixed fee for in-state students: \$2,000.
- Upper bound of fixed fee for out-of-state students: \$4,000.
- Lower bound of fixed fee for in-state students: \$500.
- Lower bound of fixed fee for out-of-state students: \$1,000.
- Max tuition per credit for in-state Engineering: \$1,333.
- Max tuition per credit for in-state Arts: \$1,000.
- Max tuition per credit for out-of-state Engineering: \$2,333.
- Max tuition per credit for out-of-state Arts: \$1,667.

^aTroutt, M. D. (1983). Deciding Tuition Structure with Linear Programming. Research in Higher Education, 18(3), 359–371. <http://www.jstor.org/stable/40195542>

Linear Programming Study Set

13/21

Correct Answer 8.

Decision variables:

x_1 : Fixed fee for in-state students (Student fees)

x_2 : Variable fee per credit for in-state Engineering students

x_3 : Variable fee per credit for in-state Arts students

x_4 : Fixed fee for out-of-state students (Student fees)

x_5 : Variable fee per credit for out-of-state Engineering students

x_6 : Variable fee per credit for out-of-state Arts students

Objective Function

$$\text{Max } Z = E_i(x_1 + Cx_2) + A_i(x_1 + Cx_3) + E_o(x_4 + Cx_5) + A_o(x_4 + Cx_6)$$

Constraints

$$E_i(x_1 + Cx_2) + A_i(x_1 + Cx_3) + E_o(x_4 + Cx_5) + A_o(x_4 + Cx_6) \geq 10,000,000 \quad (1)$$

$$\frac{E_i(x_1 + Cx_2) + A_i(x_1 + Cx_3) + E_o(x_4 + Cx_5) + A_o(x_4 + Cx_6)}{E_i + A_i + E_o + A_o} \leq 25,000 \quad (2)$$

$$500 \leq x_1 \leq 2,000 \quad (3)$$

$$x_2 \leq 1333 \quad (4)$$

$$x_3 \leq 1000 \quad (5)$$

$$1,000 \leq x_4 \leq 4,000 \quad (6)$$

$$x_5 \leq 2333 \quad (7)$$

$$x_6 \leq 1667 \quad (8)$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0 \quad (9)$$

The first constraint ensures that the total income from all students meets a minimum income requirement of \$10,000,000, to cover the operational costs of the university. The second constraint imposes an upper limit on the average tuition per student, which ensures that the tuition fees remain reasonable and do not exceed a predetermined threshold, making it a crucial constraint for maintaining affordability and accessibility. The final set of constraints specifies the lower and upper bounds for both fixed and variable fees for in-state and out-of-state students.

```
from gurobipy import Model, GRB

# Create a new model
m = Model("UniversityTuition")

# Define Parameters
E_i = 100 # Expected number of in-state Engineering students
A_i = 150 # Expected number of in-state Arts students
E_o = 150 # Expected number of out-of-state Engineering students
A_o = 100 # Expected number of out-of-state Arts students
C = 15 # Average number of credits taken by a student
```

Linear Programming Study Set

14/21

```
# Upper bounds for fixed fees
fixed_fee_ub_in_state = 2000
fixed_fee_ub_out_of_state = 4000
fixed_fee_lb_in_state = 500
fixed_fee_lb_out_of_state = 1000

average_tuition_ub = 25000

# Add variables with upper bounds for fixed fees
x1 = m.addVar(lb= fixed_fee_lb_in_state , ub=fixed_fee_ub_in_state, name="
    x1") # Fixed fee for in-state students (Student fees)
x2 = m.addVar(ub=1333, name="x2") # Variable fee per credit for in-state
    Engineering students
x3 = m.addVar(ub=1000, name="x3") # Variable fee per credit for in-state
    Arts students
x4 = m.addVar(lb= fixed_fee_lb_out_of_state, ub=fixed_fee_ub_out_of_state,
    name="x4") # Fixed fee for out-of-state students (Student fees)
x5 = m.addVar(ub=2333, name="x5") # Variable fee per credit for out-of-
    state Engineering students
x6 = m.addVar(ub=1667, name="x6") # Variable fee per credit for out-of-
    state Arts students

# Set the Objective Function
m.setObjective(E_i * (x1 + C * x2) + A_i * (x1 + C * x3) + E_o * (x4 + C *
    x5) + A_o * (x4 + C * x6), GRB.MAXIMIZE)

# Add Constraint for Minimum Income Requirement
m.addConstr(E_i * (x1 + C * x2) + A_i * (x1 + C * x3) + E_o * (x4 + C * x5)
    + A_o * (x4 + C * x6) >= 10000000, "MinimumIncomeRequirement")

# Add Constraint to maintain the average tuition below a certain upper
    bound
m.addConstr((E_i * (x1 + C * x2) + A_i * (x1 + C * x3) + E_o * (x4 + C * x5
    ) + A_o * (x4 + C * x6)) / (E_i + A_i + E_o + A_o) <= average_tuition_ub
    , "AverageTuitionUpperBound")

# Optimize model
m.optimize()

if m.status == GRB.OPTIMAL:
    print(f'Optimal_Fixed_fee_for_in-state_students:_{x1.x:.2f}')
    print(f'Optimal_Variable_fee_per_credit_for_in-state_Engineering_
        students:_{x2.x:.2f}')
    print(f'Optimal_Variable_fee_per_credit_for_in-state_Arts_students:_{x3.x:.2f}')
    print(f'Optimal_Fixed_fee_for_out-of-state_students:_{x4.x:.2f}')
    print(f'Optimal_Variable_fee_per_credit_for_out-of-state_Engineering_
        students:_{x5.x:.2f}')
    print(f'Optimal_Variable_fee_per_credit_for_out-of-state_Arts_students:
        _{x6.x:.2f}')
    print(f'Maximum_Total_Income:_{m.objVal:.2f}')
else:
    print('No_Optimal_Solution_Found')
```

Linear Programming Study Set

15/21

Question 9. Consider the LP below and write out its dual. Solve the LP and its dual via your favorite solver.

$$\begin{aligned} \text{Maximize } & Z = 10x_1 + 14x_2 + 20x_3, \\ \text{Subject to: } & 2x_1 + 3x_2 + 4x_3 \leq 220 \\ & 4x_1 + 2x_2 - x_3 \leq 385, \\ & x_1 + 4x_3 \leq 160 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

CorrectAnswer 9. The dual of the problem is given below:

$$\begin{aligned} \text{Minimize } & W = 220y_1 + 385y_2 + 160y_3, \\ \text{Subject to: } & 2y_1 + 4y_2 + y_3 \geq 10 \\ & 3y_1 + 2y_2 \geq 14 \\ & 4y_1 - y_2 + 4y_3 \geq 20 \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

```
# Create the primal model
m = Model("Primal")

# Create variables
x1 = m.addVar(vtype=GRB.CONTINUOUS, name="x1")
x2 = m.addVar(vtype=GRB.CONTINUOUS, name="x2")
x3 = m.addVar(vtype=GRB.CONTINUOUS, name="x3")

# Set objective
m.setObjective(10*x1 + 14*x2 + 20*x3, GRB.MAXIMIZE)

# Add constraints
m.addConstr(2*x1 + 3*x2 + 4*x3 <= 220, "c0")
m.addConstr(4*x1 + 2*x2 - x3 <= 385, "c1")
m.addConstr(x1 + 4*x3 <= 160, "c2")

# Optimize model
m.optimize()

# Print results
if m.status == GRB.OPTIMAL:
    print('Optimal solution found for Primal:')
    for v in m.getVars():
        print('%s: %g' % (v.varName, v.x))
    print('Objective Value: %g' % m.objVal)
```

Linear Programming Study Set

16/21

```
# Create the dual of the model
md = Model("Dual")

# Create variables
y1 = md.addVar(vtype=GRB.CONTINUOUS, name="y1")
y2 = md.addVar(vtype=GRB.CONTINUOUS, name="y2")
y3 = md.addVar(vtype=GRB.CONTINUOUS, name="y3")

# Set objective
md.setObjective(220*y1 + 385*y2 + 160*y3, GRB.MINIMIZE)

# Add constraints
md.addConstr(2*y1 + 4*y2 + y3 >= 10, "c0")
md.addConstr(3*y1 + 2*y2 >= 14, "c1")
md.addConstr(4*y1 - y2 + 4*y3 >= 20, "c2")

# Optimize model
md.optimize()

# Print results
if md.status == GRB.OPTIMAL:
    print('Optimal solution found for Dual:')
    for v in md.getVars():
        print('%s: %g' % (v.varName, v.x))
    print('Objective Value: %g' % md.objVal)
```


Linear Programming Study Set

17/21

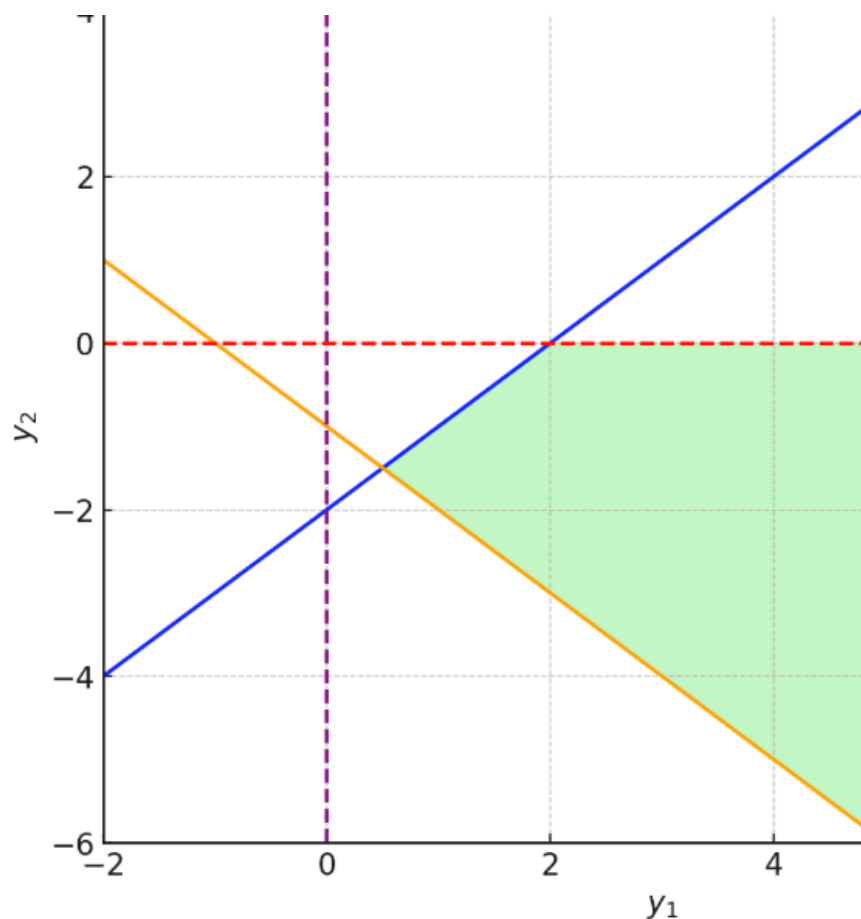
Question 10. Find the dual of the following LP and use the dual solution to show the primal LP is infeasible.

$$\begin{aligned} \text{Maximize } & Z = 2x_1 - x_2 \\ \text{Subject to: } & x_1 + x_2 \leq 2 \\ & -x_1 + x_2 \geq 6, \\ & x_1, x_2 \geq 0 \end{aligned}$$

CorrectAnswer 10. The dual of the problem is given below:

$$\begin{aligned} \text{Minimize } & W = 2y_1 + 6y_2, \\ \text{Subject to: } & y_1 - y_2 \geq 2 \\ & y_1 + y_2 \geq -1, \\ & y_1 \geq 0, \\ & y_2 \leq 0 \end{aligned}$$

We draw the dual problem using the graphical solution method, as seen in Figure below. The dual of the LP is unbounded; thus, the primal LP must be infeasible. (we used the weak duality theorem to find this)



Linear Programming Study Set

18/21

Question 11. *Candy Kane Cosmetics (CKC) produces Leslie Perfume, which requires chemicals and labor. Two production processes are available: Process 1 transforms 1 unit of labor and 2 units of chemicals into 3 oz of perfume. Process 2 transforms 2 units of labor and 3 units of chemicals into 5 oz of perfume. It costs CKC \$3 to purchase a unit of labor and \$2 to purchase a unit of chemicals. Each year, up to 20,000 units of labor and 35,000 units of chemicals can be purchased. In the absence of advertising, CKC believes it can sell 1,000 oz of perfume. To stimulate demand for Leslie, CKC can hire a promoter, who is paid \$100/hour. Each hour the promoter works for the company is estimated to increase the demand for Leslie Perfume by 200 oz. Each ounce of Leslie Perfume sells for \$5.*

1. *Formulate the problem as an LP and use a software of your choice to compute the optimal solution.*
2. *What if the estimated increase in demand with the promoter was 250 oz instead of by 200 oz. How would the optimal solution change?*
3. *What if there was another promoter, who is paid \$75/hour and their associated estimated increase in demand for the company was by 110 oz. How would the optimal solution change?*

Linear Programming Study Set

19/21

Correct Answer 11.

12.1. Let:

- x_1 : Number of units of Process 1.
- x_2 : Number of units of Process 2.
- x_3 : Number of hours to hire the promoter.

Objective Function:

To maximize the profit:

$$\text{Maximize } Z = 8x_1 + 13x_2 - 100x_3 \quad (10)$$

Constraints:

Subject to the following constraints:

$$x_1 + 2x_2 \leq 20000 \quad (\text{Labor Constraint}) \quad (11)$$

$$2x_1 + 3x_2 \leq 35000 \quad (\text{Chemical Constraint}) \quad (12)$$

$$3x_1 + 5x_2 - 200x_3 \leq 1000 \quad (\text{Demand Constraint}) \quad (13)$$

$$x_1, x_2, x_3 \geq 0 \quad (\text{Non-negativity Constraint}) \quad (14)$$

12.2. If the estimated increase in demand with the promoter changes from 200 oz to 250 oz per hour, it would affect the demand constraint in the model. The new constraint will look like this:

$$3x_1 + 5x_2 - 250x_3 \leq 1000$$

12.3. When introducing a new promoter, you'll have an additional decision variable and constraints to consider, based on which promoter(s) you choose to hire.

Let's denote:

- x_4 : Number of hours to hire the new promoter.

Modified Objective Function: The cost associated with the new promoter would be subtracted in the objective function.

$$\text{Maximize } Z = 8x_1 + 13x_2 - 100x_3 - 75x_4 \quad (15)$$

Modified and Additional Constraints: The demand constraint would need to be updated to incorporate the increased demand due to the second promoter.

$$3x_1 + 5x_2 - 200x_3 - 110x_4 \leq 1000$$

Linear Programming Study Set

20/21

12.1.

```
m = Model("CKC")

# Add decision variables
x1 = m.addVar(vtype=GRB.CONTINUOUS, name="x1")
x2 = m.addVar(vtype=GRB.CONTINUOUS, name="x2")
x3 = m.addVar(vtype=GRB.CONTINUOUS, name="x3")

# Integrate new variables
m.update()

# Set the objective function
m.setObjective(8 * x1 + 13 * x2 - 100 * x3, GRB.MAXIMIZE)

# Add constraints
m.addConstr(x1 + 2 * x2 <= 20000, "labor_constraint")
m.addConstr(2 * x1 + 3 * x2 <= 35000, "chemical_constraint")
m.addConstr(3 * x1 + 5 * x2 - 200 * x3 <= 1000, "demand_constraint")

# Optimize the model
m.optimize()

# Display the optimal solution
if m.status == GRB.OPTIMAL:
    print('Optimal solution found!')
    for v in m.getVars():
        print('%s: %g' % (v.varName, v.x))
    print('Objective: %g' % m.objVal)
```

12.2.

```
# Remove the old demand constraint
m.remove(m.getConstrByName("demand_constraint"))

# Add the updated demand constraint
m.addConstr(3 * x1 + 5 * x2 <= 1000 + 250 * x3, "demand_constraint")

# Re-optimize the model with the updated constraint
m.optimize()

# Display the updated optimal solution
if m.status == GRB.OPTIMAL:
    print('Optimal solution found!')
    for v in m.getVars():
        print('%s: %g' % (v.varName, v.x))
    print('Objective: %g' % m.objVal)
```

Linear Programming Study Set

21/21

12.3.

```
# Add the new decision variable for the second promoter
x4 = m.addVar(vtype=GRB.CONTINUOUS, name="x4")

# Integrate the new variable
m.update()

# Modify the Objective Function to include the cost associated with the new
  promoter
m.setObjective(m.getObjective() - 75 * x4, GRB.MAXIMIZE) # Note the
  subtraction of the new term

# Modify the Demand Constraint to include the additional demand generated
  by the new promoter
demand_constr = m.getConstrByName("demand_constraint")
m.chgCoeff(demand_constr, x4, 110) # 110 being the increase in demand by
  the new promoter

# Re-optimize the model with the updated objective function and constraint
m.optimize()

# Display the updated optimal solution
if m.status == GRB.OPTIMAL:
    print('Updated Optimal solution found!')
    for v in m.getVars():
        print('%s: %g' % (v.varName, v.x))
    print('Updated Objective: %g' % m.objVal)
```